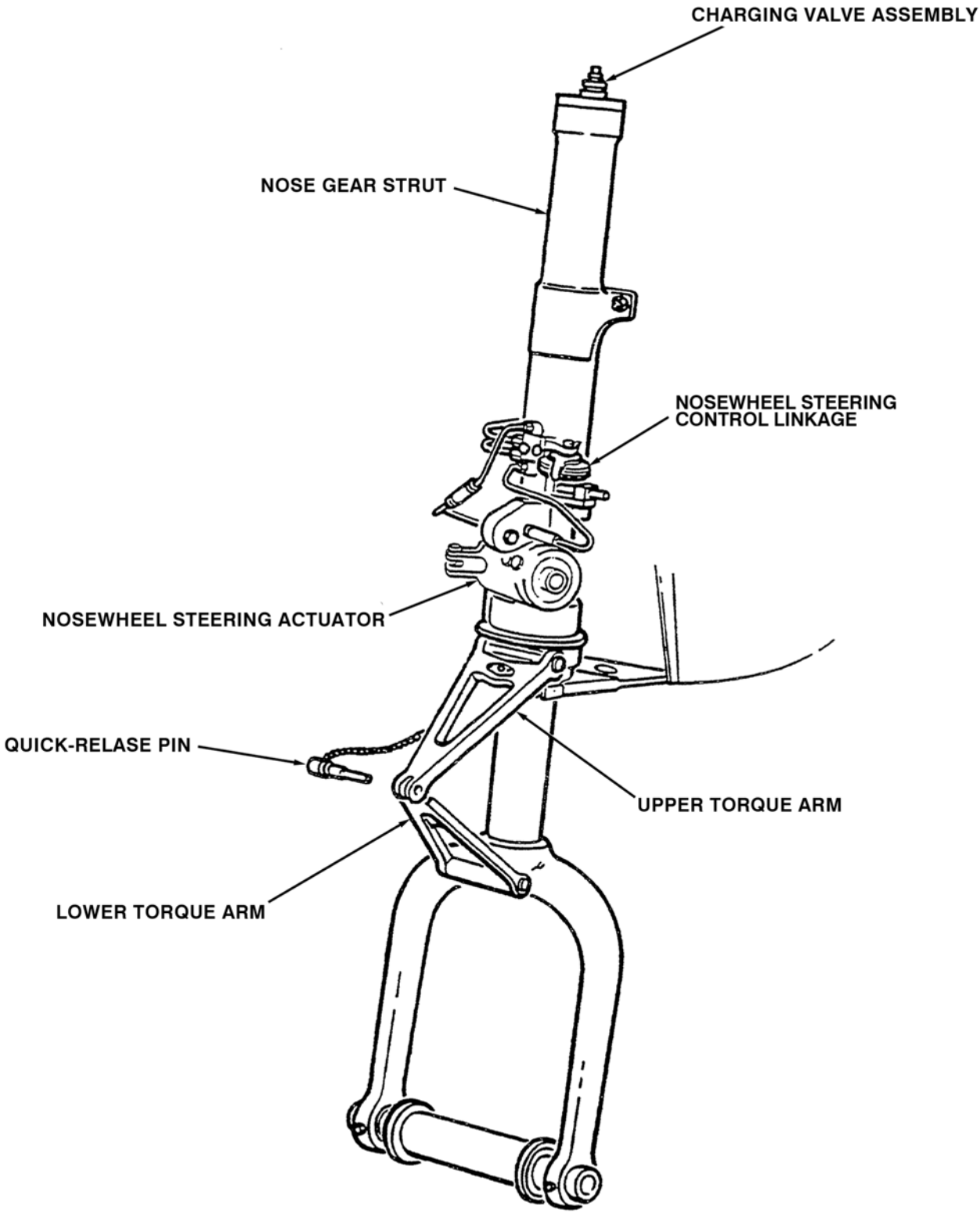
Nose Gear Shock Strut – General Data

1. General

Refer to Figure 1 for details.

The nose gear shock strut consists of an outer cylinder, piston assembly, nose wheel fork and a steering collar. The cylinder forms the main structure of the strut and provides the mounting for the nose wheel steering actuator (refer to 32-50-00, Nose Wheel Steering System – General Data). The nose wheel fork is secured to the piston and telescopes within the cylinder which is charged with oil and air, damping being achieved by restricting the oil and air flow through orifices in the piston. A valve, through which the strut may be recharged, is located on top of the strut.



Nose Gear Shock Strut
Figure 1

2. Servicing

A. Charge Nose Gear Shock Strut

The nose gear strut must be serviced in accordance with the instructions on the nose gear identification plate.

Note Dry compressed nitrogen is preferred to compressed air for charging the shock strut.

3. Removal/Installation

Refer to Figure 2 for details.

A. Remove Nose Gear Strut

CAUTION ENSURE ALL AIR PRESSURE IS RELEASED BEFORE REMOVING NOSE WHEEL FORK FROM NOSE GEAR SHOCK STRUT.

1) Open nose baggage compartment. Remove screens and covers as required.

2) Jack aircraft until nose wheel is clear of ground. (Refer to 07-00-00, Lifting and Shoring – General Data).

3) Discharge hydraulic system accumulator pressure by operating wing flaps.

4) Disconnect hydraulic lines at swivel fittings. Cap open ports and lines.

5) Disconnect nose wheel steering cables at turnbuckles.

6) Support strut and remove two lower attachment bolts and cap fitting. Remove two upper attachment bolts.

B. Install Nose Gear Strut

1) Position nose gear strut on forward face of nose compartment bulkhead and install two upper attachment bolts. Install cap fitting and two lower attachment bolts.

2) Torque strut attachment bolts to 250 to 300 pound–inches, and ensure attachment bolts have 0.02 inch minimum clearance between end of bolt and side of hole. Ensure cap fitting has 0.03 inch minimum gap on each side. Lockwire bolts.

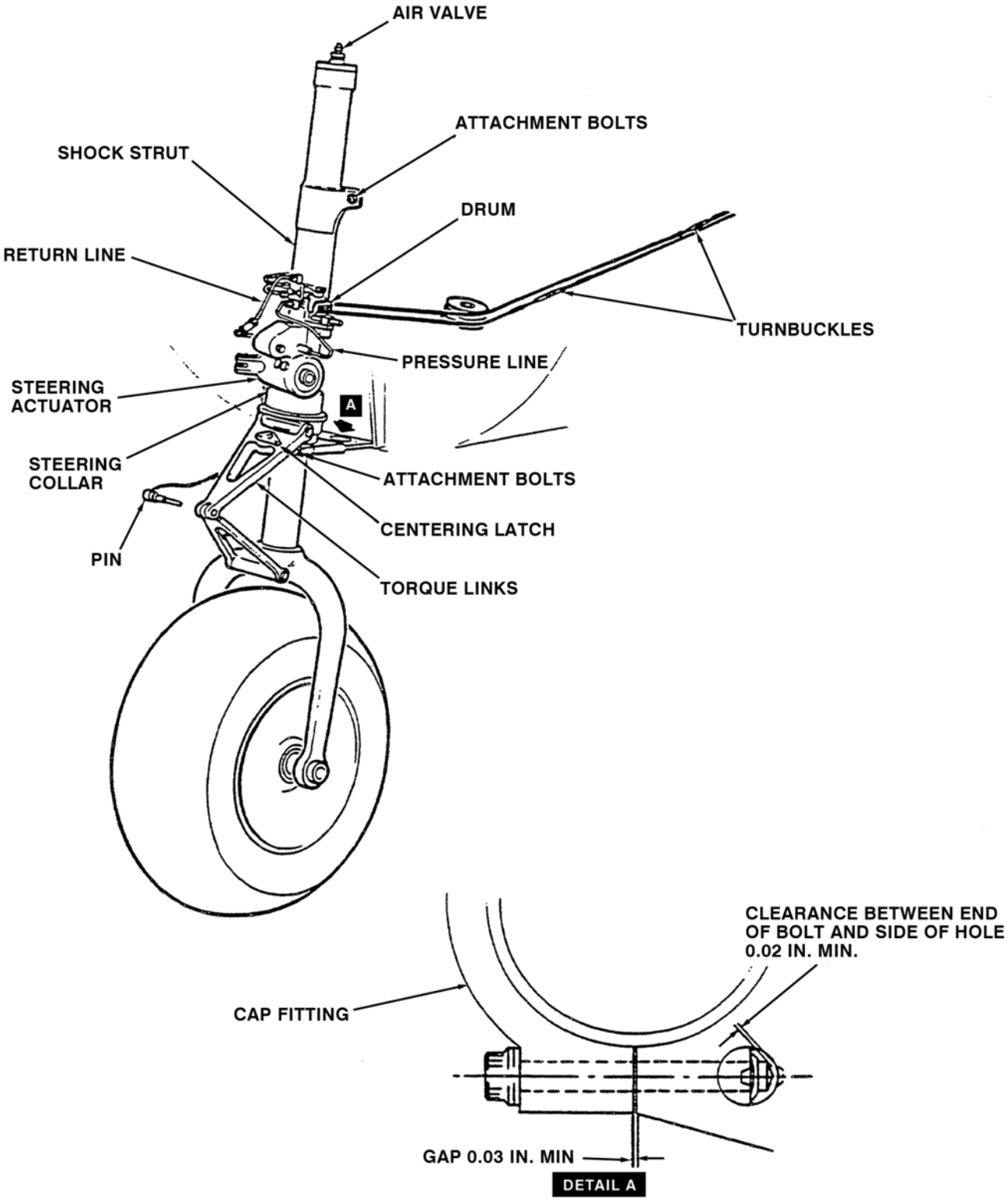
3) Connect nose wheel steering cables at turnbuckles.

4) Connect hydraulic lines at swivel fittings.

5) Bleed nose wheel steering system. (Refer to 32-50-00, Nose Wheel Steering System – General Data).

6) Rig and function test nose wheel steering system. (Refer to 32-50-00, Nose Wheel Steering System – General Data).

7) Install screens and covers as necessary. Close nose baggage compartment.



Nose Gear Shock Strut Installation
Figure 2

4. Adjustment/Test

A. Pressure Test

CAUTION POUR HYDRAULIC FLUID. DO NOT FILL WITH HYDRAULIC FLUID UNDER PRESSURE.

1) Jack nose wheel clear of ground. Remove valve cap and release air slowly.

2) Remove air charging valve and slowly compress strut fully to expel hydraulic fluid.

Note If no hydraulic fluid is expelled, the strut is almost empty. Add approximately 10 fluid ounces (300 cc) of hydraulic fluid, Specification MIL–H–5606 and repeat Para 4.A.2).

3) Extend strut fully with valve open to permit air to enter strut. Ensure ground clearance with strut fully extended. Compress strut to expel additional oil.

4) Repeat Para 4.A.3) five times until no additional oil is expelled.

5) Add 25 fluid ounces (725 cc) of hydraulic fluid, Specification MIL–H–5606, while slowly extending strut.

6) Install pressure gauge, and with strut fully extended, inflate with dry compressed nitrogen or dry compressed air to 95 psi. There must be no pressure loss after 48 hours.

7) Ensure there is no sign of fluid leakage at upper or lower ends of cylinder, or at joint of piston and fork.

8) Remove pressure gauge and install valve cap. Lower aircraft to ground.

5. Inspection/Check

A. Wear Check-Nose Gear Torque Links

1) Remove quick-release pin and separate upper and lower torque links.

2) Move upper torque link so that one lug is tight against face of steering collar; using feeler gauges, check gap between other lug and face on steering collar. Gap must not exceed 0.008 inch.

3) Repeat Para 5.A.2) and check for wear on lower torque link.

4) Check for wear between lug of lower torque link and two lugs of upper torque link. Gap must not exceed 0.008 inch.

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